Jhalak Upadhyay

Backend & Platform Engineer

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SUMMARY

Backend engineer with 2+ years of experience building distributed systems, event pipelines, and cloud platforms in Java and TypeScript. Shipped GPU-fleet observability and compute-provisioning platforms at early-stage startups. Google Summer of Code mentor and Badge Magic repository maintainer at FOSSASIA.

EXPERIENCE

Velerion (spun off from Substrate.ai) · SDE, Backend & Platform — Remote · Feb 2026 – Present

Velerion · Jun 2026 – Present (moved with the founding team at the spin-off)

- Built a GPU-fleet observability platform end to end (Bun + Hono API, BullMQ workers on Valkey, ClickHouse rollups), ingesting telemetry from a self-updating DCGM agent across 60+ hosts and 300+ GPUs.
- Designed an alerting pipeline over GPU and host signals (thermals, VRAM, ECC/XID errors, heartbeats) with sustained-duration debounce and grouped notifications, cutting duplicate alerts by ~70%.
- Built the backend for the enterprise section of the GPU cloud console, covering volumes, images, SSH keys, object storage, networking, firewalls, and load balancers.
- Delivered the operator dashboard (React 19, TanStack Router) on these APIs, with fleet overview, per-host drill-down, metrics explorer, logs, alerts, and an audit trail.

Substrate.ai · Feb 2026 – Jun 2026

- Built a provider-agnostic GPU compute marketplace that unifies the Gcore and Shadeform APIs behind one interface for reserved and on-demand capacity, cutting provisioning time from hours to minutes.
- Enforced multi-tenant data isolation on Supabase (Postgres, Auth, Edge Functions) through row-level security policies, hardened authentication, and per-tenant rate limiting.
- Built Grafana dashboards for real-time GPU utilization, job health, and billing metrics, used by SRE for incident response and by customers for usage reporting.

Gpu.net · Software Development Engineer I — Bangalore · Jun 2025 – Feb 2026

- Designed an event-driven fan-out feed service on a pub/sub event bus that aggregates events from 3 products into a personalized per-user feed, load-tested at 1M+ events/day.
- Built Spring Boot REST APIs with input validation, JWT authentication, and rate limiting, load-tested to 1,000 requests/sec at peak.
- Reduced API response time by ~40% through PostgreSQL indexing, query tuning, and a Redis caching layer.
- Led development of the Gpunet consumer app, integrating the Astra agent marketplace, the feed service, and the Gvex trading platform.

Infinisync Consulting · Software Development Engineer I — Delhi · May 2024 – May 2025

Full-time during final year of B.Tech. under a college NOC (No Objection Certificate)

- Built a backend that automated lead generation and appointment scheduling, reducing the client sales team's manual work by 60%.
- Engineered a Java orchestration pipeline that synchronizes generative AI audio with avatar video, using Apache Kafka for asynchronous chunked streaming to achieve frame-accurate lip-sync.
- Wrote unit and integration tests with JUnit and Mockito, and set up CI pipelines with GitHub Actions and Maven.

TECHNICAL SKILLS

Languages: Java, TypeScript, SQL
Backend: Spring Boot, Node.js (Bun, Hono), REST APIs, JWT Authentication, Rate Limiting, Event-Driven Architecture (Pub/Sub, Fan-out)
Messaging & Queues: Apache Kafka, BullMQ
Databases & Storage: PostgreSQL, ClickHouse, Redis / Valkey, Supabase (Auth, Row-Level Security, Edge Functions), Object Storage
Observability & Infrastructure: Grafana, NVIDIA DCGM, Alerting Pipelines, GPU Cloud Provisioning (Gcore, Shadeform), Load Balancers, Firewalls
Testing & CI/CD: JUnit, Mockito, Integration Testing, Load Testing, GitHub Actions, Maven
Frontend & Mobile (working knowledge): React 19, TanStack Router, Flutter

OPEN SOURCE

Google Summer of Code, FOSSASIA · Mentor & Maintainer (Mentee in 2024) — Jun 2024 – Present

- Built the Badge Magic Flutter app from scratch as a GSoC 2024 mentee, replacing legacy native iOS and Android apps and shipping cross-platform BLE transfer to LED badges (3K+ Play Store downloads, 1.9K+ GitHub stars).
- Returned as a mentor to guide international contributors through scoping, architecture, and code review, and to plan releases with the FOSSASIA core team.
- Serve as a maintainer of the Badge Magic repository with admin access: triage issues, review and merge contributor pull requests, enforce code quality standards, and manage releases.

EDUCATION

KIET Group of Institutions · B.Tech. in Computer Science and Engineering — 2021 – 2025 · CGPA 8.3/10